AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 11 · STUDENT LESSON

Updating parameters

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Save the old state. Update together. Check the new loss.

Apply an ordered gradient and learning rate while auditing state provenance and overshoot.

Keep one habit: Complete same-start rate table → pictured target5update → mixed-state comparison → zero/invalid probes → independent new model.

MAIN LESSON · SECTION 1

Did every parameter update use the same old state?

> Save the gradient,apply the rate,then recompute the actual loss.

Learning goals

I can apply a **learning rate** to an ordered gradient.

I can update all parameters from the same saved state.

I can verify the new loss and identify **overshoot**ing.

Retrieve the ordered gradient

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- For x = 2, target 3, w = 1 and b = 0, what is the squared-loss gradient in order [w, b]?

> **My starting explanation**

End goal

By the end, I can explain updating parameters using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Apply a **learning rate** to an ordered gradient

Update all parameters from the same saved state

Verify the new loss and identify **overshoot**ing

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Subtract the rate times each matching gradient entry

> A negative gradient increases its parameter under this rule.

Declare the update and complete original example

Gradient descent uses new parameter = old parameter - **learning rate** times its gradient entry. The minus sign chooses the locally downhill direction for a positive **learning rate**.

A negative gradient therefore increases the corresponding parameter.

Keep the Week 10 example: x = 2, target 3, y = 2w + b and L = (y - 3) squared. At [w, b] = [1, 0], the loss is 1 and gradient is [-4, -2].

With **learning rate** 0.1, new w = 1 - 0.1 times -4 = 1.4 and new b = 0 - 0.1 times -2 = 0.2.

Complete before-and-after record

Quantity	Before update	After update
Parameters [w, b]	[1, 0]	[1.4, 0.2]
Prediction	2	3

Quantity	Before update	After update
Squared loss	1	0

Limit the one-example success claim

This particular step reaches zero loss on one example. It does not establish a unique parameter solution, successful learning on other examples or a generally ideal **learning rate**.

Name given and missing facts

List the given facts and one fact that is not supplied.

MAIN LESSON · SECTION 3

Compare four complete one-step experiments from the same start

> Hold the original parameters,gradient,data example and loss fixed while changing rate.

Complete rate/prediction/loss table

Fresh start[w, b]=[1, 0]for every row · Saved gradient · New[w, b] · New prediction 2w+b · New loss(target 3)

> **Fresh start[w, b]=[1, 0]for every row:** Rate0

Saved gradient: [−4, −2]

New[w, b]: [1, 0]

New prediction 2w+b: 2

New loss(target 3): 1

Fresh start[w, b]=[1, 0]for every row: Rate.05

Saved gradient: [−4, −2]

New[w, b]: [1.2, .1]

New prediction 2w+b: 2.5

New loss(target 3): 0.25

Fresh start[w, b]=[1, 0]for every row: Rate.1

Saved gradient: [−4, −2]

New[w, b]: [1.4, .2]

New prediction 2w+b: 3

New loss(target 3): 0

Fresh start[w, b]=[1, 0]for every row: Rate.3

Saved gradient: [−4, −2]

New[w, b]: [2.2, .6]

New prediction 2w+b: 5

New loss(target 3): 4

Restart,simultaneous-state and helper contract

These are four separate one-step experiments, not four consecutive updates. Each restarts from[1, 0]and computes both gradient entries before either parameter changes. Rate0is a no-change comparison; positive rates choose the negative-gradient direction, but a finite step can overshoot. The function is specific to x2, target 3 and assumes numeric inputs.

It does not validate a rate, select one automatically or train on multiple examples. Distinct parameter pairs can also fit one example:both[1.4, .2]and[1, 1]predict 3.

Predict the rows: unchanged parameters and loss 1 at rate 0; [1.2, 0.1] and loss 0.25 at rate 0.05; [1.4, 0.2] and loss 0 at rate 0.1; [2.2, 0.6] and loss 4 at rate 0.3.

The helper performs one step on this fixed example; it assumes numeric **input**s and does not choose a **learning rate** for you.

Runnable one-step implementation

```
def step(w, b, rate):
    residual = 2 * w + b - 3
    dw, db = 4 * residual, 2 * residual
    new_w = w - rate * dw
    new_b = b - rate * db
    new_loss = (2 * new_w + new_b - 3) ** 2
    return new_w, new_b, new_loss

for rate in [0.0, 0.05, 0.1, 0.3]:
    print(rate, tuple(round(v, 6)
                      for v in step(1, 0, rate)))
```

Connect the picture and parameter values

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Connect the picture and the explanation

Change the rate while keeping the gradient experiment fixed

> The direction alone does not settle the resulting loss.

Current overshoot task

- For the lesson start [1, 0], apply rate 0.3 and calculate **prediction** and loss.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured larger rate

- Change: Set **learning rate** 0.2 from the same starting point.

▼ **Check after predicting**

Prediction: New w=2.6, b=1.8, y=7, squared loss 4: no improvement.

Reason: The larger update jumps to an equally distant point on the opposite side of the target.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

Trace a full update and its checked consequence

> Keep the pictured target5separate from the main target3example.

Complete picture facts and separate-model scope

Use one explicit gradient update and then recompute the loss.

Model y=2w+b, target 5. Start w=1, b=1; **learning rate** 0.05. Initial y=3, error y−target=−2, squared loss 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

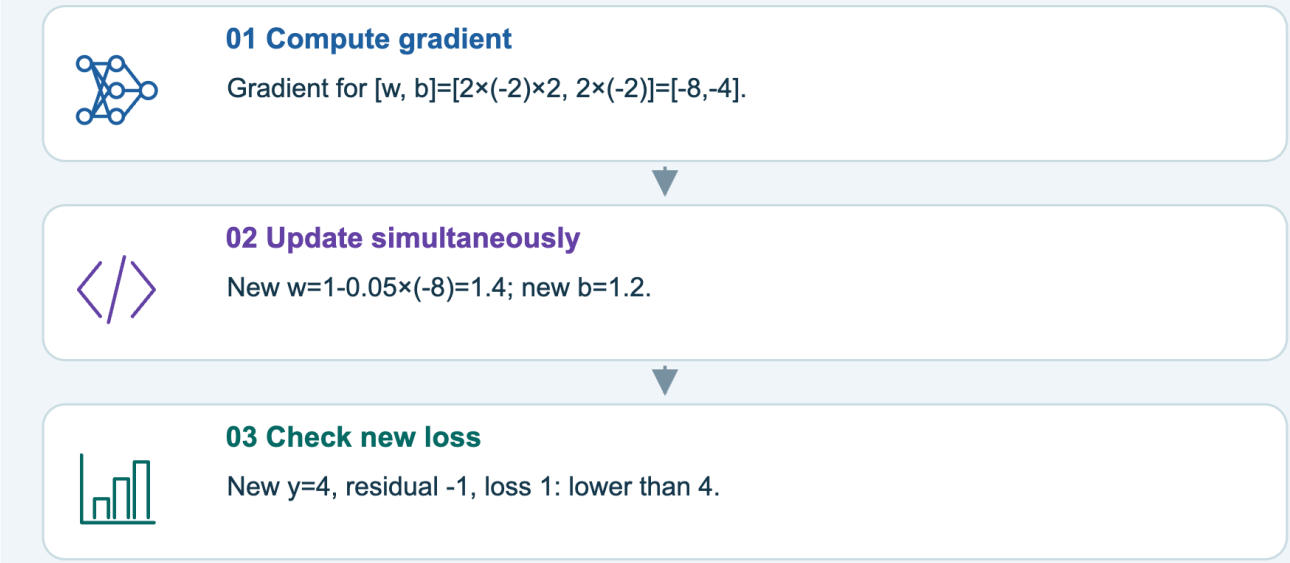
The picture changes the target and start: x2, target 5, start[w, b]=[1, 1], prediction 3, residual−2, loss 4, gradient[−8, −4]. Rate.05 gives[1.4, 1.2], prediction 4, loss 1. Restarting from that original start with rate.2 gives[2.6, 1.8], prediction 7, loss 4:an equally distant overshoot. Do not use the main target 3 gradient[−4, −2]in this fixture.

Both cases compute all new values from the same old state and check the new loss.

Trace gradient,simultaneous update and new loss

AI ACADEMY / AI-5/11 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute gradient	Gradient for [w, b]=[2×(-2)×2, 2×(-2)]=[-8,-4].
2. Update simultaneously	New w=1-0.05×(-8)=1.4; new b=1.2.
3. Check new loss	New y=4, residual -1, loss 1: lower than 4.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

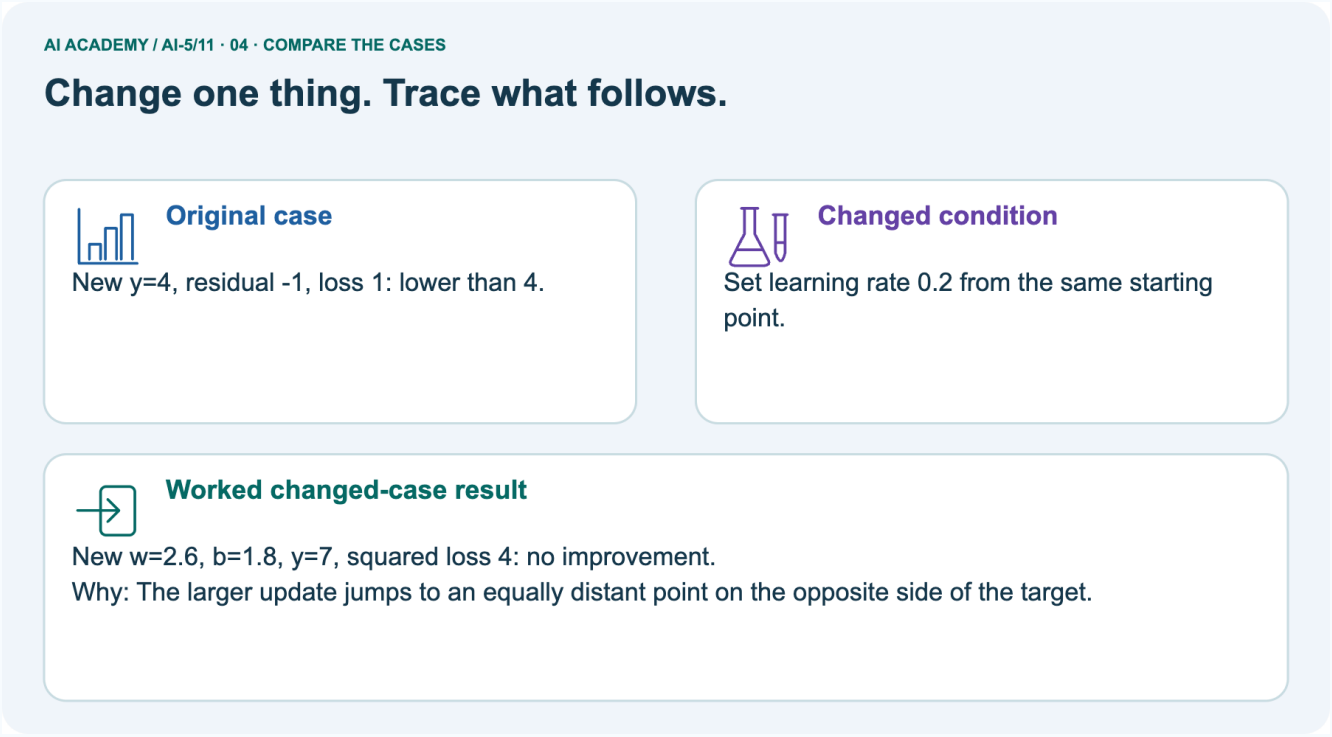
Gradient descent subtracts **learning rate** times gradient. A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original [w,b] before updating either one.

The small step brings the **prediction** closer to five, reducing squared error. A larger step can cross the target and undo that gain, so the new **prediction** and loss must be recalculated.

Predict the larger-rate restart

- Change one thing: Set **learning rate** 0.2 from the same starting point. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain overshoot and remaining evidence limits

- One successful toy update is not **evidence** of convergence for a real network or generalisation to unseen data.

- Read the labelled picture: Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try the main rate.05update

Use one gradient for one parameter snapshot

STAGE	SAVED VALUES	ACTION
Before the step	w is 1 and b is 0	Compute gradient [-4, -2]
Apply rate 0.1	Use both old parameters	Produce [1.4, 0.2]
After the step	Use both new parameters	Recompute loss 0

Read top to bottom. All gradient entries come from the same old model. The final row evaluates the fully updated model rather than mixing old and new coordinates.

Starting again from [1, 0], use learning rate 0.05. Show both new parameters, the new prediction and the new loss.

▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

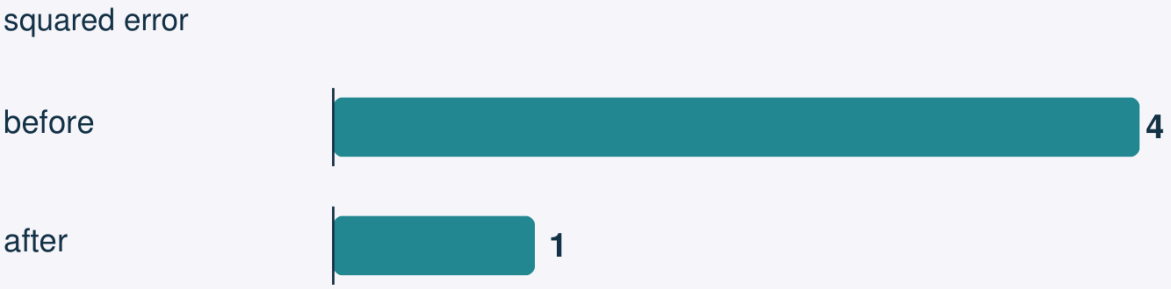
Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Updating parameters

The facts we will use

AI ACADEMY / AI-5/11

Loss before and after rate 0.05 update



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/11

Updating parameters

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gradient

Gradient for [w, b]=[2 x (-2) x 2, 2 x (-2)]=[-8,-4].



2. Update simultaneously

New w=1-0.05 x (-8)=1.4; new b=1.2.



3. Check new loss

New y=4, residual -1, loss 1: lower than 4.

CHANGE ONE THING

Set learning rate 0.2 from the same starting point.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Repair a mixed-state update even when its loss falls

> Numerical improvement is not proof that the specified algorithm was implemented.

Complete mixed-state explanation and overshoot

Suppose a program first changes w to 1.4, then recomputes the bias gradient using new w and old b. Its residual is now -0.2, so its bias gradient is -0.4 and new b becomes 0.04. **Prediction** becomes 2.84 and loss 0.0256.

That is a different procedure from the declared simultaneous step, even though it also lowered this loss.

An excessive rate can increase loss. At rate 0.3, both coordinates move in the locally suggested direction but **prediction** jumps to 5 and loss rises from 1 to 4. Always evaluate and record the result.

Later lessons compare learning-rate behaviour across repeated updates.

Same-start side-by-side state audit

Main start[1, 0], rate.1	Saved simultaneous procedure	Mixed-state procedure
w gradient evaluated at old[1, 0]	−4	−4
New w	1.4	1.4
b gradient comes from	Old[1, 0]:−2	New w1.4/old b0:−.4
New b	.2	.04
New prediction /loss	3 /0	2.84 /.0256
Meaning	Matches the declared simultaneous update	Different update procedure despite lower loss

Explain the first procedure mismatch

- Explain why recomputing the bias gradient after changing w is not the **simultaneous update** specified here.

> **State one repair and the checks you would rerun.**

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise signed updates and state provenance with less help

> Retain the original parameters and gradient beside the new values.

Current task and response

- Starting with parameter a = 2, gradient -3 and **learning rate** 0.2, calculate the new a.

> **Starting again from [1, 0], use learning rate 0.05. Show both new parameters, the new prediction and the new loss.**

> **Explain your prediction to a partner before comparing results.**

Supported W2record

Use workbook W2 for the connected example “Updating parameters”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for Updating parameters. Proposed explanation: “Subtracting a negative gradient decreases the parameter”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured checked-loss reasoning

- Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Reserve the independent new model

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Detect overshooting

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate rate,state and the actual post-update loss

> The complete probes distinguish algorithm correctness from an observed decrease.

Current task and runnable one-step helper

For the lesson start [1, 0], apply rate 0.3 and calculate **prediction** and loss.

```
def step(w, b, rate):
    residual = 2 * w + b - 3
    dw, db = 4 * residual, 2 * residual
    new_w = w - rate * dw
    new_b = b - rate * db
    new_loss = (2 * new_w + new_b - 3) ** 2
    return new_w, new_b, new_loss
```

```
for rate in [0.0, 0.05, 0.1, 0.3]:
    print(rate, tuple(round(v, 6)
                      for v in step(1, 0, rate)))
```

Complete normal,zero,overshoot,missing and mixed-state probes

Complete update probe	Expected answer / action
Normal main rate.05	New[1.2, .1], prediction 2.5, loss.25
No-change rate 0	Original[1, 0], prediction 2, loss 1
Overshoot rate.3	New[2.2, .6], prediction 5, loss 4
Zero-gradient start[1, 1]	Prediction3, loss 0; gradient[0, 0], parameters unchanged
Missing rateNone	Outside numeric contract; request/validate before calling
Rate−.1in the helper	New[.6, −.2], prediction 1, loss 4; negative-rate arithmetic is not the declared positive-rate descent
Recomputed b gradient after new w	Gives mixed-state[1.4, .04], not simultaneous[1.4, .2]
Lower training loss after a step	Evidence about these training cases/objective only; no untouched final or generalization claim

Predict,restart,run or trace,and check

1. Predict every supplied one-step result and restart the same original parameters for each rate. Run or trace the helper, record the saved gradient, both new parameters, new prediction and checked loss. Compare the complete mixed-state record with the simultaneous result. Use separate runs for invalid inputs; the helper assumes numeric values and is not a rate validator.

   No final-test references should enter training gradients. A zero gradient leaves this update unchanged but is not a global-minimum certificate for arbitrary objectives.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: update a different input and target

> Recompute its gradient from its supplied original state before applying the rate.

Complete new independent model and update

Use x = 1, target 4, w = 1, b = 1, squared loss and **learning rate** 0.1. Compute **prediction**, residual, gradient [2 times residual times x, 2 times residual], new parameters and new loss.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Update a new model independently

Attempt this without hints

Use x = 1, target 4, w = 1, b = 1, squared loss and **learning rate** 0.1. Compute **prediction**, residual, gradient [2 times residual times x, 2 times residual], new parameters and new loss.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep one complete update record with its source state

> The project ledger should show exactly which values produced the gradient.

Project artifact and evidence table

Add a labelled updating parameters **evidence** sheet to Image-pattern Network Investigation. Show how you can apply a **learning rate** to an ordered gradient.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one state audit and one checked loss

> Explain why a finite downhill-directed step still needs evaluation.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Apply a **learning rate** to an ordered gradient

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Learning rate	A multiplier that controls the size of a gradient update
Gradient descent	Updating parameters in the direction opposite the gradient
Simultaneous update	Computing all new parameter values from the same old state
Overshoot	A step that moves too far and can increase the objective

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Starting with parameter a = 2, gradient -3 and **learning rate** 0.2, calculate the new a.
- Explain why recomputing the bias gradient after changing w is not the **simultaneous update** specified here.
- At a point with gradient [0, 0], what does this update do? Does that alone prove a global minimum on any loss surface?

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent x1,target4simultaneous update for the first individual attempt.

Current explanation and response

- A lower loss after a step supports progress on the objective and examples used for that step. It does not show that a model is safe, calibrated or better on unseen data.

  Keep the data split fixed and never use final-test labels to compute **training** gradients.

- Next week repeats the forward, loss, gradient and update cycle in a small network with a clearly stated stopping budget.

- At a point with gradient [0, 0], what does this update do? Does that alone prove a global minimum on any loss surface?

A teammate says any update that lowers loss is a correct implementation of simultaneous gradient descent. Use the mixed-state example to evaluate that claim.

My limited claim and a useful next check

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Use one gradient for one parameter snapshot

STAGE	SAVED VALUES	ACTION
Before the step	w is 1 and b is 0	Compute gradient [-4, -2]
Apply rate 0.1	Use both old parameters	Produce [1.4, 0.2]
After the step	Use both new parameters	Recompute loss 0

Read top to bottom. All gradient entries come from the same old model. The final row evaluates the fully updated model rather than mixing old and new coordinates.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Use one explicit gradient update and then recompute the loss.

By the end, I can	How I will show it
Apply a learning rate to an ordered gradient	Explain the deciding step, show a trace or test, and state one limit.
Update all parameters from the same saved state	Explain the deciding step, show a trace or test, and state one limit.
Verify the new loss and identify overshooting	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Use one explicit gradient update and then recompute the loss.
- Which labelled step matches this detail? New y=4, residual -1, loss 1: lower than 4.
- 1. Check new loss
- 2. Compute gradient
- 3. Update simultaneously
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check new loss New y=4, residual -1, loss 1: lower than 4. **Gradient descent** subtracts **learning rate** times gradient.

A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original [w,b] before updating either one. The small step brings the **prediction** closer to five, reducing squared error.

A larger step can cross the target and undo that gain, so the new **prediction** and loss must be recalculated.

Changed-case question and optional hint

- Changed-case practice: Set **learning rate** 0.2 from the same starting point. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. New w=2.6, b=1.8, y=7, squared loss 4: no improvement.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: New w=2.6, b=1.8, y=7, squared loss 4: no improvement. The larger update jumps to an equally distant point on the opposite side of the target.

Repair and retain the record

Review Gradients with a numerical check (ai-5/10). Goal: Estimate one partial derivative while other parameters stay fixed. Short practice: Why should the numerical checks perturb one coordinate at a time?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Loss functions (ai-5/8). Goal: Calculate per-example squared error and mean squared error. Short practice: Why must the report name the chosen loss?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Composing layers (ai-5/5). Goal: Trace how one layer supplies the next layer **input**s. Short practice: What intermediate value explains the changed **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

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<summary>Reference · Six-picture concept journey</summary>

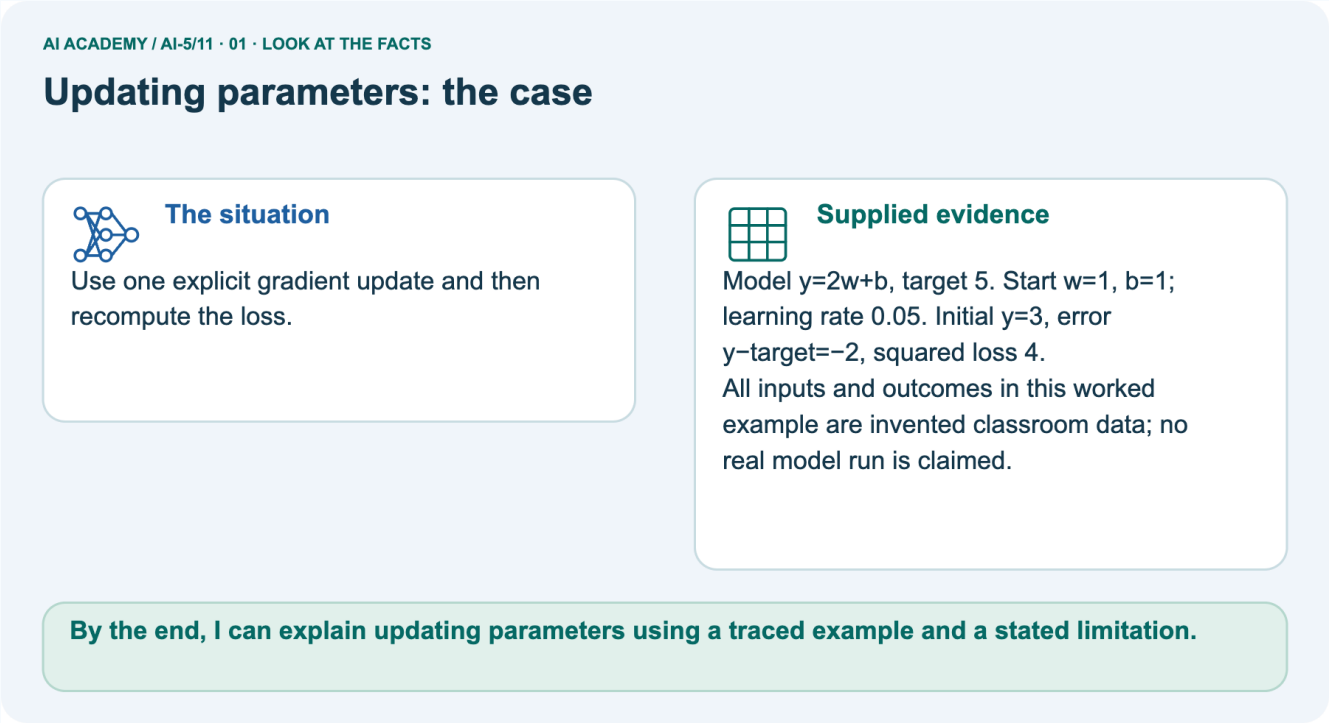
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REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Updating parameters: the case



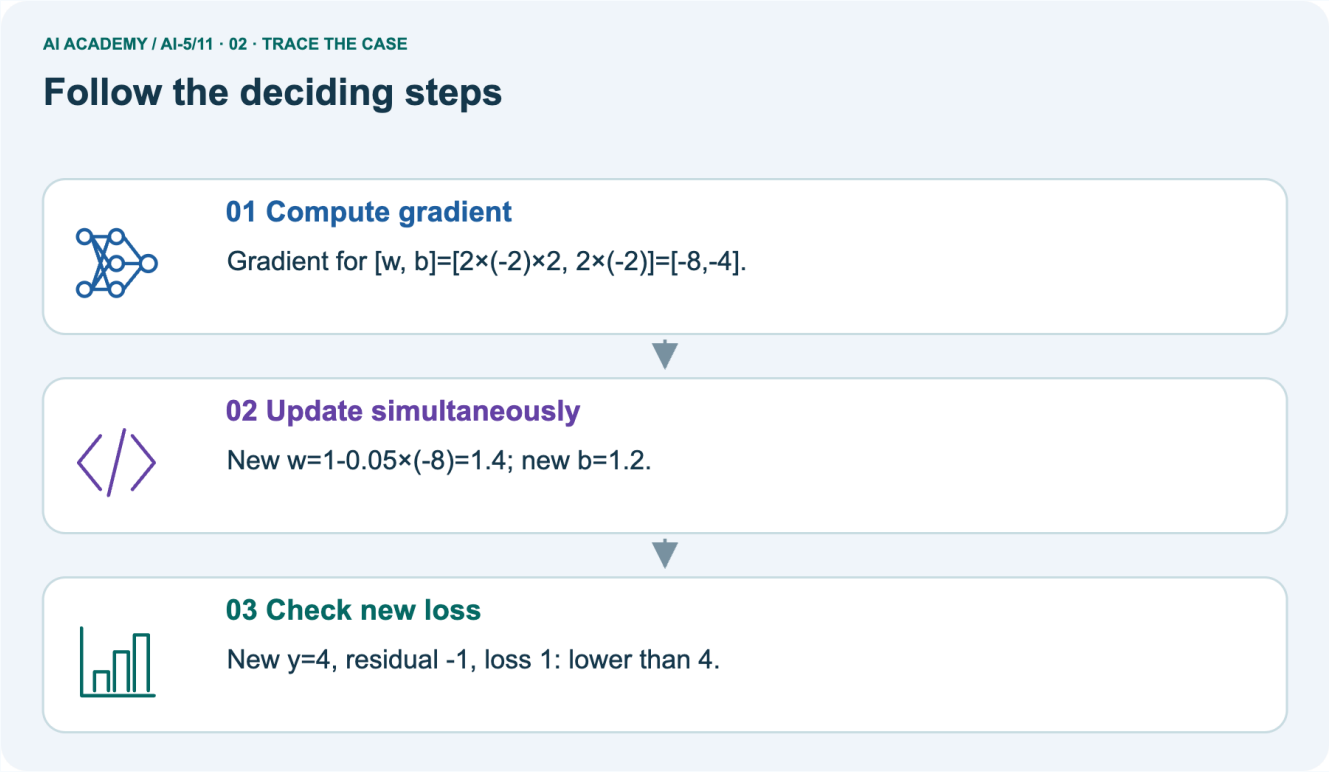
Follow the labelled picture with the matching explanation. Open the vector diagram

Use one explicit gradient update and then recompute the loss. Supplied facts: Model y=2w+b, target 5. Start w=1, b=1; **learning rate** 0.05. Initial y=3, error y−target=−2, squared loss 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

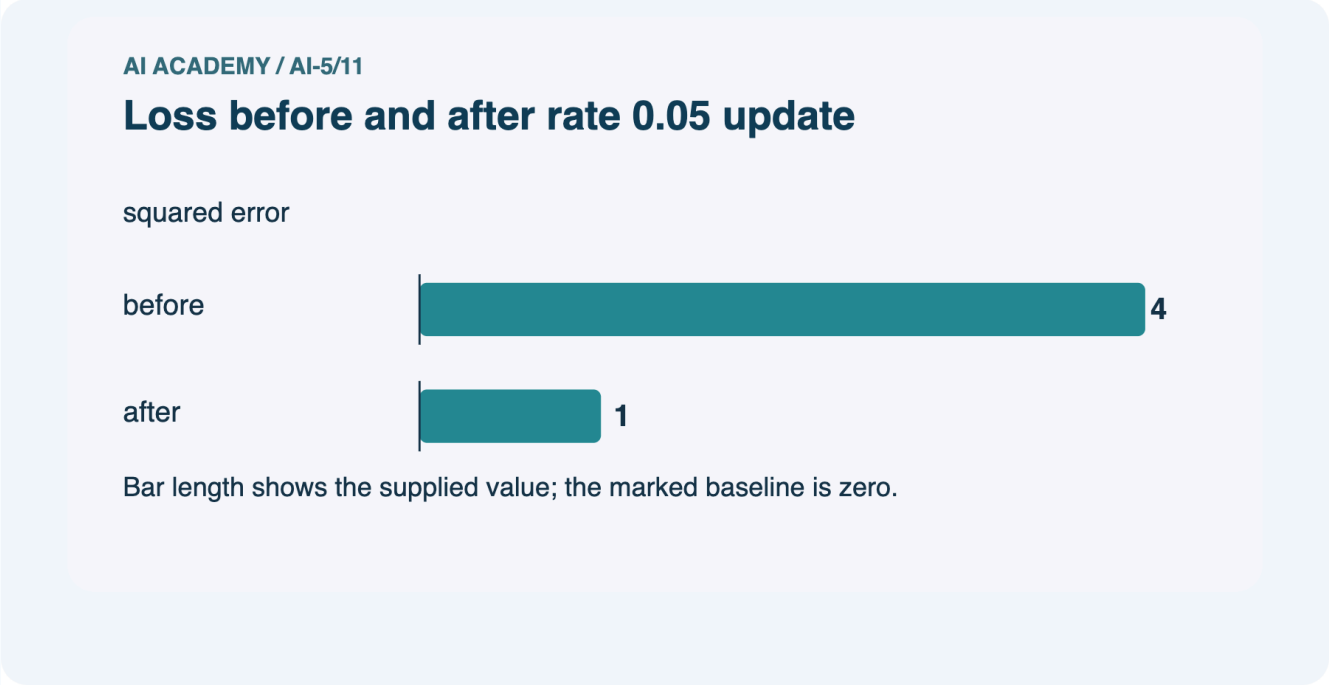


Follow the labelled picture with the matching explanation. Open the vector diagram

Compute gradient: Gradient for [w, b]=[2×(-2)×2, 2×(-2)]=[-8,-4]. → Update simultaneously: New w=1-0.05×(-8)=1.4; new b=1.2. → Check new loss: New y=4, residual -1, loss 1: lower than 4.

Point to each numbered stage and explain which recorded **input** it uses.

03. Loss before and after rate 0.05 update

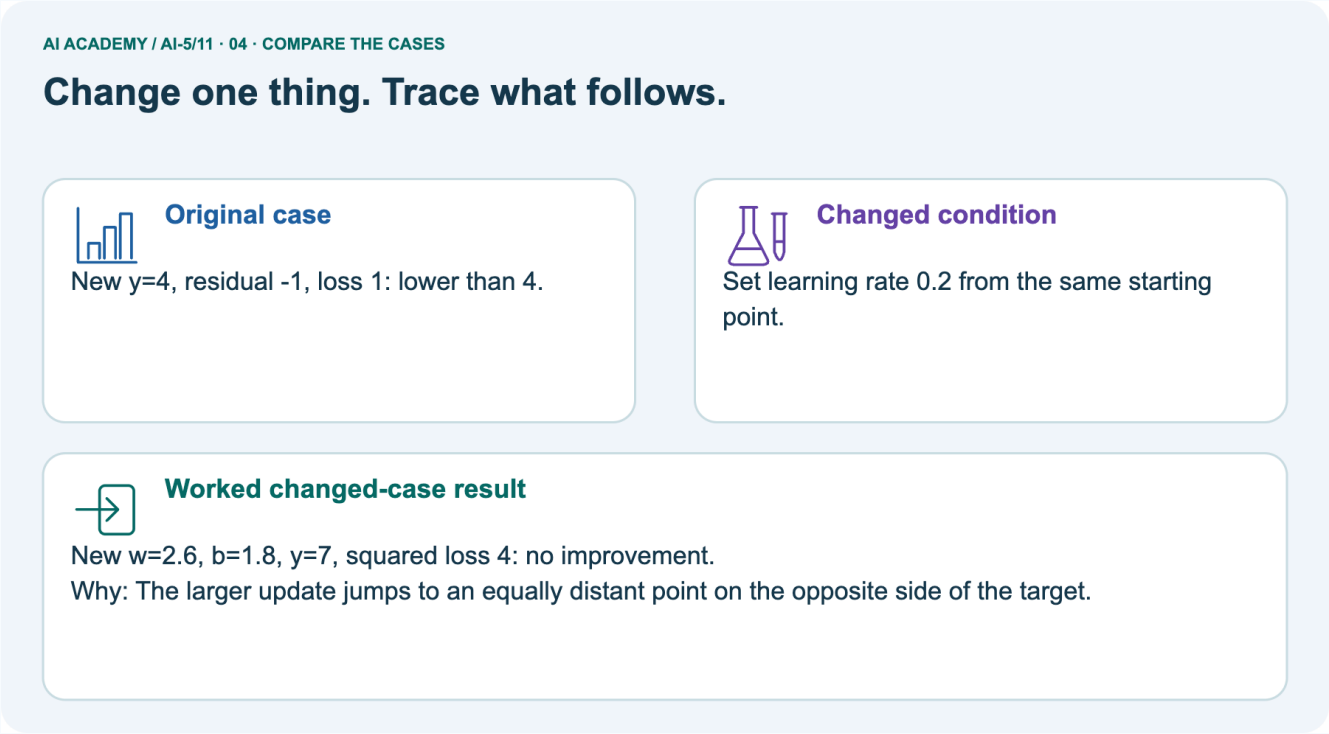


Follow the labelled picture with the matching explanation. Open the vector diagram

Loss before and after rate 0.05 update. Units: squared error. before: 4. after: 1.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



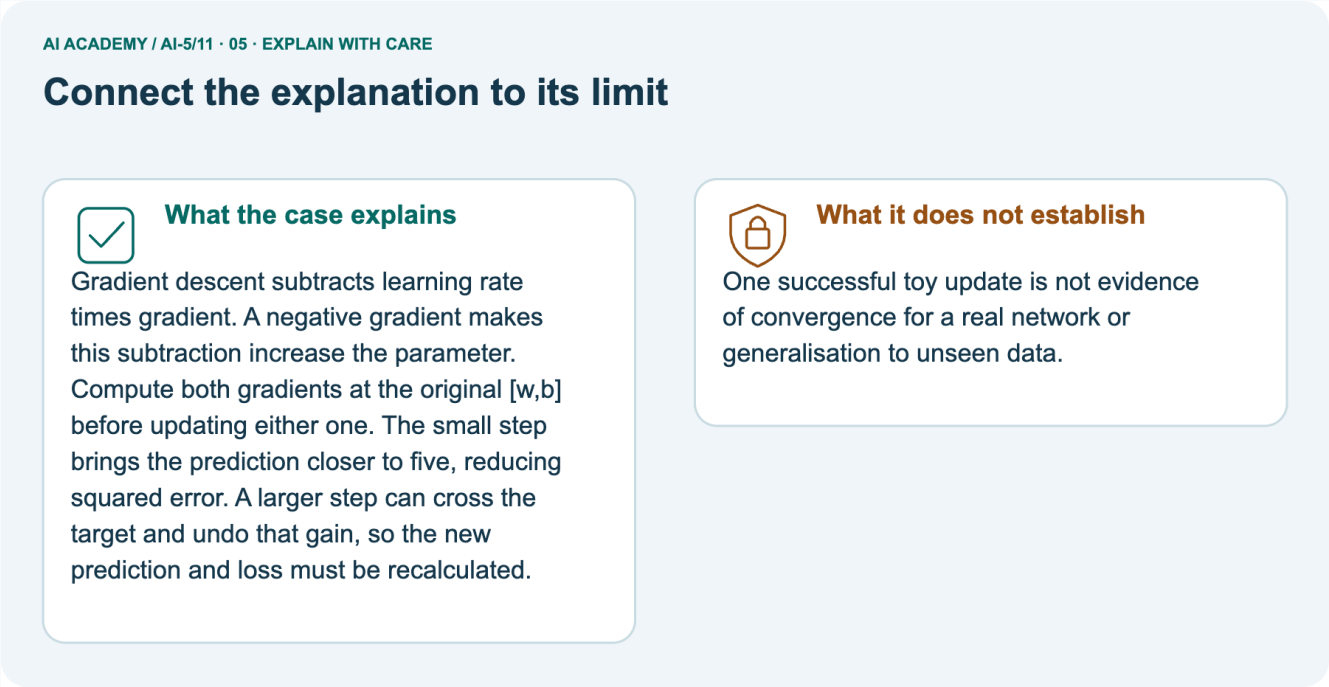
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: New y=4, residual -1, loss 1: lower than 4. Changed condition: Set **learning rate** 0.2 from the same starting point. Expected result: New w=2.6, b=1.8, y=7, squared loss 4: no improvement.

Reason: The larger update jumps to an equally distant point on the opposite side of the target.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

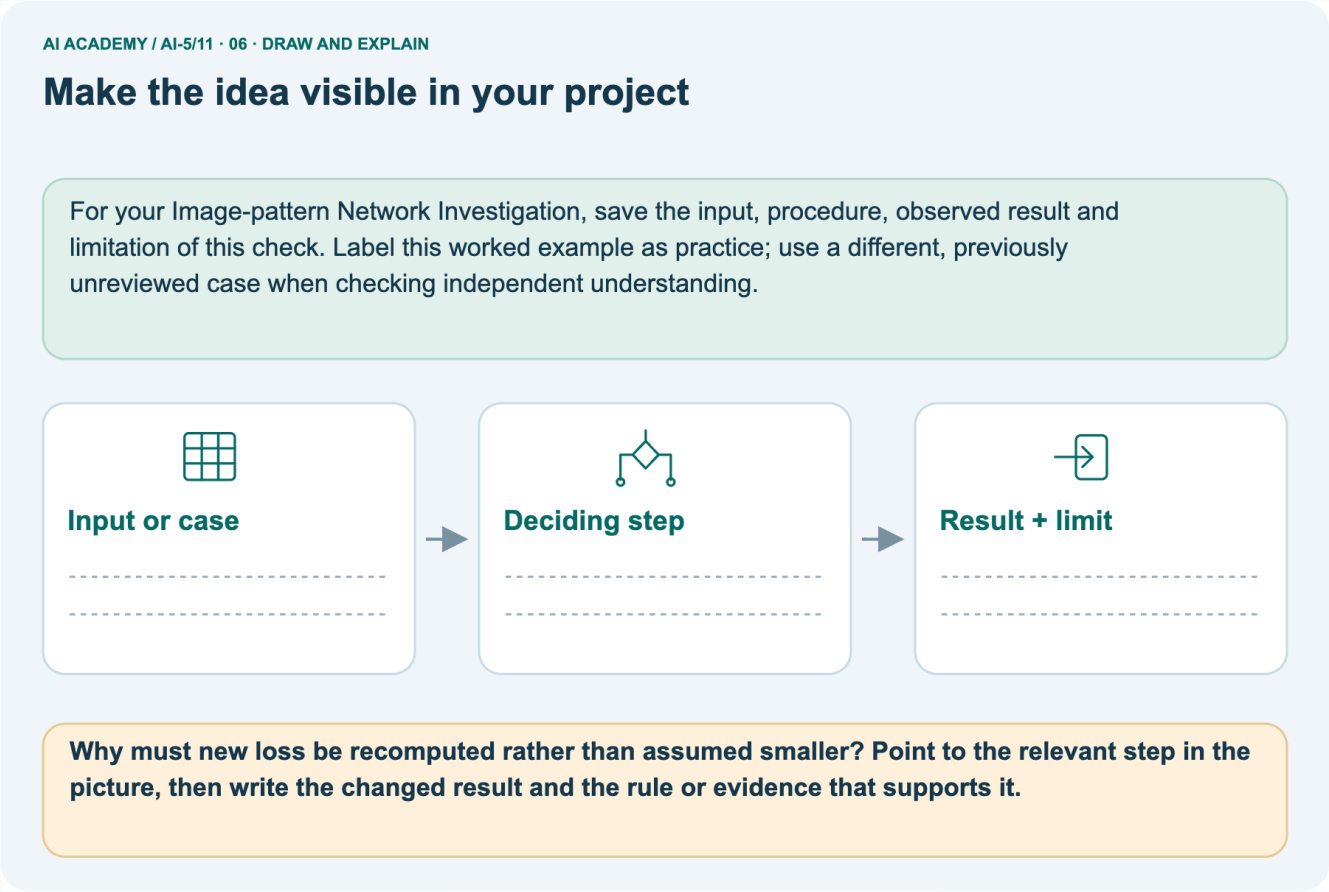
Gradient descent subtracts **learning rate** times gradient. A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original [w,b] before updating either one.

The small step brings the **prediction** closer to five, reducing squared error. A larger step can cross the target and undo that gain, so the new **prediction** and loss must be recalculated.

Limit: One successful toy update is not **evidence** of convergence for a real network or generalisation to unseen data.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Updating parameters: the case

02. Follow the deciding steps

03. Loss before and after rate 0.05 update

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.